Multi-View Semantic 3D Gaussian Splatting for Component-Aware Bridge Reconstruction from UAV Imagery

Rendering bridge appearance and structural components from arbitrary viewpoints

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IC-SHM Competition – Project 2
September 2026



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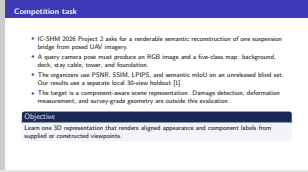
Good morning. I am Quoc-Bao Ta, presenting joint work with Trung Ha Nguyen and Thai Ha Dang. We develop a multi-view semantic Gaussian Splatting system that reconstructs bridge appearance and structural component labels from UAV images.

Competition task

- IC-SHM 2026 Project 2 asks for a renderable semantic reconstruction of one suspension bridge from posed UAV imagery.
- A query camera pose must produce an RGB image and a five-class map: background, deck, stay cable, tower, and foundation.
- The organizers use PSNR, SSIM, LPIPS, and semantic mIoU on an unreleased blind set. Our results use a separate local 30-view holdout [1].
- The target is a component-aware scene representation. Damage detection, deformation measurement, and survey-grade geometry are outside this evaluation.

Objective

Learn one 3D representation that renders aligned appearance and component labels from supplied or constructed viewpoints.



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Project 2 asks us to render both an RGB image and a five-class semantic map from a query camera. The official blind evaluation combines visual and semantic metrics. Our aim is therefore one scene representation that keeps appearance and component labels aligned. We do not claim damage detection or survey-grade geometry.

Released data and required output

- 400 UAV photographs at 1320×989 pixels.
- 300 images contain polygons for four bridge components and background.
- 100 images contain RGB only. Our pipeline generates pseudo-labels so these views can provide semantic supervision.
- SIMPLE_RADIAL calibration, camera poses, and COLMAP feature tracks are supplied, but no precomputed 3D points.

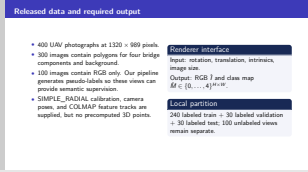
Renderer interface

Input: rotation, translation, intrinsics, image size.

Output: RGB $\hat{I}$ and class map $\hat{M} \in \{0, \ldots, 4\}^{H \times W}$.

Local partition

240 labeled train + 30 labeled validation + 30 labeled test; 100 unlabeled views remain separate.



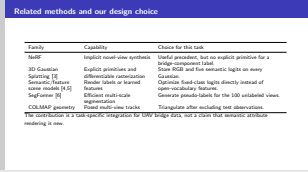
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The released data contain 400 posed UAV images at 1320 by 989 pixels. Three hundred images have component polygons, while one hundred contain RGB only. We divide the labeled set into 240 training, 30 validation, and 30 local test images. The renderer receives camera parameters and returns RGB plus a five-class map.

Related methods and our design choice

Family	Capability	Choice for this task
NeRF	Implicit novel-view synthesis	Useful precedent, but no explicit primitive for a bridge-component label.
3D Gaussian Splatting [3]	Explicit primitives and differentiable rasterization	Store RGB and five semantic logits on every Gaussian.
Semantic/feature scene models [4,5]	Render labels or learned features	Optimize fixed-class logits directly instead of open-vocabulary features.
SegFormer [6]	Efficient multi-scale segmentation	Generate pseudo-labels for the 100 unlabeled views.
COLMAP geometry	Posed multi-view tracks	Triangulate after excluding test observations.

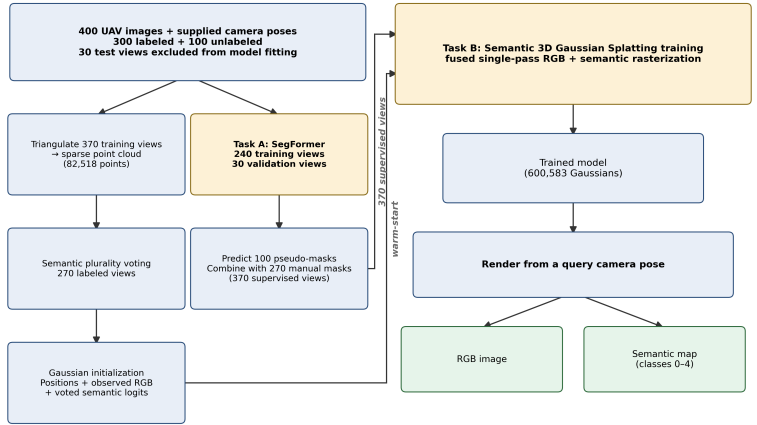
The contribution is a task-specific integration for UAV bridge data, not a claim that semantic attribute rendering is new.



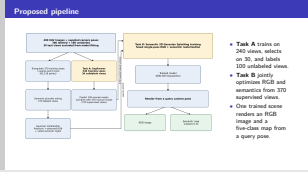
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Our design combines established tools for this specific bridge task. COLMAP tracks provide sparse geometry, SegFormer supplies missing semantic supervision, and 3D Gaussian Splatting provides an explicit differentiable renderer. Our contribution is the integrated RGB and five-logit representation, rather than a claim that semantic feature rendering itself is new.

Proposed pipeline



- **Task A** trains on 240 views, selects on 30, and labels 100 unlabeled views.
- **Task B** jointly optimizes RGB and semantics from 370 supervised views.
- One trained scene renders an RGB image and a five-class map from a query pose.



Target: 0:25.

The pipeline has two tasks. Task A fine-tunes SegFormer on 240 labeled views, selects the checkpoint on 30 validation views, and predicts masks for the 100 unlabeled views. Task B combines 270 manual masks with those 100 pseudo-masks. It initializes a semantic Gaussian scene from sparse geometry, then jointly optimizes appearance and semantics.

Data split and test isolation

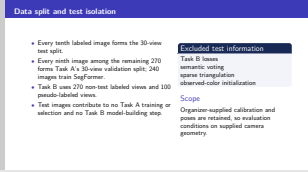
- Every tenth labeled image forms the 30-view test split.
- Every ninth image among the remaining 270 forms Task A's 30-view validation split; 240 images train SegFormer.
- Task B uses 270 non-test labeled views and 100 pseudo-labeled views.
- Test images contribute to no Task A training or selection and no Task B model-building step.

Excluded test information

Task B losses
semantic voting
sparse triangulation
observed-color initialization

Scope

Organizer-supplied calibration and poses are retained, so evaluation conditions on supplied camera geometry.



Target: 0:20.

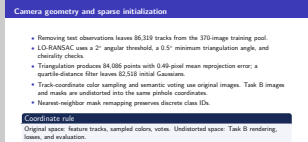
Test isolation is important. Every tenth labeled image forms the 30-view test split. The remaining labeled images provide Task A training and validation, and Task B uses all 270 non-test labeled views. Test observations do not enter losses, semantic voting, triangulation, or color initialization. From the test views, we retain the organizer-supplied camera geometry for evaluation.

Camera geometry and sparse initialization

- Removing test observations leaves 86,319 tracks from the 370-image training pool.
- LO-RANSAC uses a 2° angular threshold, a 0.5° minimum triangulation angle, and cheirality checks.
- Triangulation produces 84,086 points with 0.49-pixel mean reprojection error; a quartile-distance filter leaves 82,518 initial Gaussians.
- Track-coordinate color sampling and semantic voting use original images. Task B images and masks are undistorted into the same pinhole coordinates.
- Nearest-neighbor mask remapping preserves discrete class IDs.

Coordinate rule

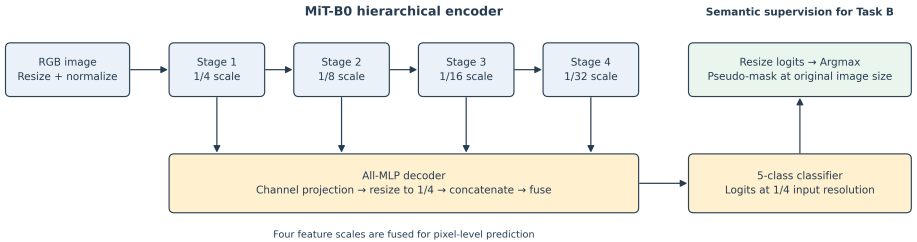
Original space: feature tracks, sampled colors, votes. Undistorted space: Task B rendering, losses, and evaluation.



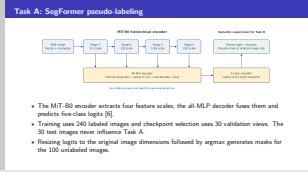
Target: 0:25.

After removing test observations, 86,319 feature tracks remain. Robust triangulation produces 84,086 points with a mean reprojection error of 0.49 pixels. Outlier filtering leaves 82,518 initial Gaussians. Tracks, color samples, and semantic votes stay in original image coordinates. Rendering and loss computation use consistently undistorted images and masks.

Task A: SegFormer pseudo-labeling



- The MiT-B0 encoder extracts four feature scales; the all-MLP decoder fuses them and predicts five-class logits [6].
- Training uses 240 labeled images and checkpoint selection uses 30 validation views. The 30 test images never influence Task A.
- Resizing logits to the original image dimensions followed by argmax generates masks for the 100 unlabeled images.



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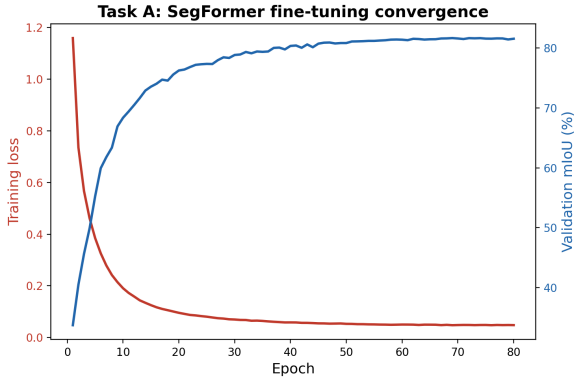
Task A uses SegFormer with the MiT-B0 encoder. Its four feature scales are fused by the all-MLP decoder to produce five-class logits. We train on 240 images and select the best checkpoint on 30 validation images. After resizing logits back to the original dimensions, argmax produces pseudo-masks for the 100 RGB-only views.

Task A protocol and validation result

Class	IoU
deck	92.80%
stay_cable	89.17%
tower	78.20%
foundation	66.48%
background	98.64%
Structural mIoU	**81.67%**

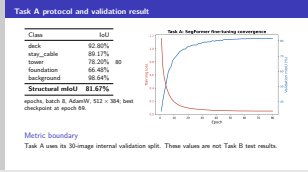
80

epochs, batch 8, AdamW, 512 × 384; best checkpoint at epoch 69.



Metric boundary

Task A uses its 30-image internal validation split. These values are not Task B test results.



Target: 0:20.

On the internal validation split, Task A reaches 81.67 percent structural mIoU, averaging the four bridge classes and excluding background. Deck and cable perform best, while foundation is the hardest class. The best checkpoint occurs at epoch 69. These values measure checkpoint selection only and are separate from the Task B test results.

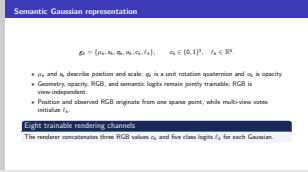
Semantic Gaussian representation

$$g_k = \{\mu_k, s_k, q_k, o_k, c_k, \ell_k\}, \qquad c_k \in (0,1)^3, \quad \ell_k \in \mathbb{R}^5.$$

- μ_k and s_k describe position and scale; q_k is a unit rotation quaternion and o_k is opacity.
- Geometry, opacity, RGB, and semantic logits remain jointly trainable; RGB is view-independent.
- Position and observed RGB originate from one sparse point, while multi-view votes initialize ℓ_k.

Eight trainable rendering channels

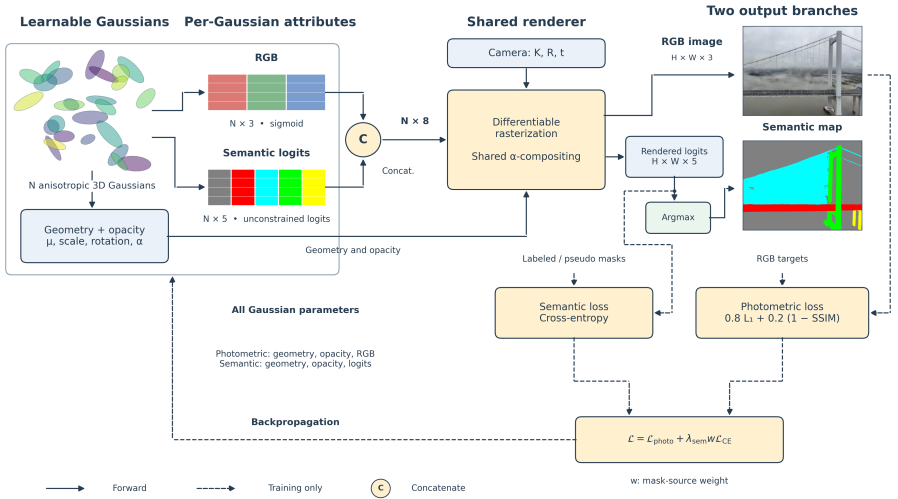
The renderer concatenates three RGB values c_k and five class logits ℓ_k for each Gaussian.

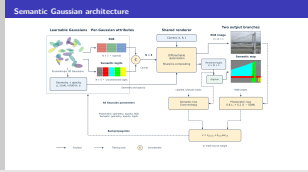


Target: 0:20.

Each Gaussian stores position, scale, rotation, opacity, three RGB values, and five semantic logits. All parameters remain trainable. Position and initial RGB come from one sparse point, while multi-view voting initializes the semantic logits. The renderer therefore processes eight trainable output channels for every Gaussian.

Semantic Gaussian architecture

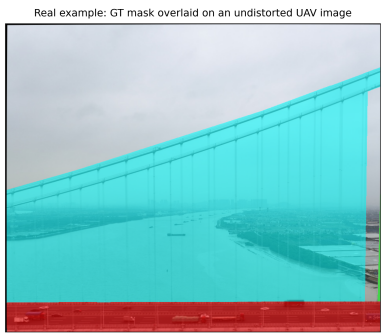


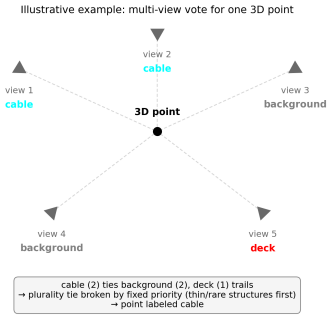


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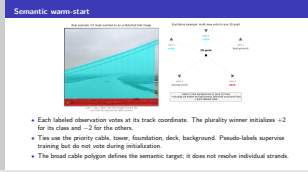
This diagram shows the complete architecture. We concatenate RGB and semantic logits before one differentiable rasterization pass. The shared projection, depth order, opacity, and alpha weights produce two aligned outputs. RGB supervision drives the photometric branch, semantic masks drive cross-entropy, and both losses update the common Gaussian representation.

Semantic warm-start





- Each labeled observation votes at its track coordinate. The plurality winner initializes $+2$ for its class and -2 for the others.
- Ties use the priority cable, tower, foundation, deck, background. Pseudo-labels supervise training but do not vote during initialization.
- The broad cable polygon defines the semantic target; it does not resolve individual strands.



Target: 0:20.

For semantic initialization, each labeled observation votes for the class of its 3D track. The plurality winner receives logit plus two, and the other classes receive minus two. A fixed priority resolves ties. Pseudo-labels supervise later optimization but do not participate in this initial vote, so initialization depends only on manual annotations.

Fused RGB and semantic rendering

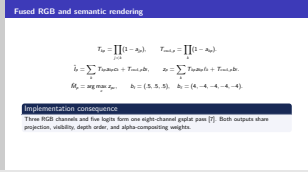
$$T_{kp} = \prod_{j<k}(1 - a_{jp}), \qquad T_{\mathrm{end},p} = \prod_{k}(1 - a_{kp}).$$

$$\hat{I}_p = \sum_k T_{kp} a_{kp} c_k + T_{\mathrm{end},p} b_I, \qquad z_p = \sum_k T_{kp} a_{kp} \ell_k + T_{\mathrm{end},p} b_\ell.$$

$$\hat{M}_p = \arg\max_c z_{pc}, \qquad b_I = (.5, .5, .5), \quad b_\ell = (4, -4, -4, -4, -4).$$

Implementation consequence

Three RGB channels and five logits form one eight-channel gsplat pass [7]. Both outputs share projection, visibility, depth order, and alpha-compositing weights.



Target: 0:20.

The renderer applies the same transmittance and alpha weight to RGB and semantic logits. Three RGB channels and five logits are concatenated into one gsplat call. Background values complete both outputs. We take argmax only after compositing the logits, which preserves spatial alignment between appearance and the evaluated semantic map.

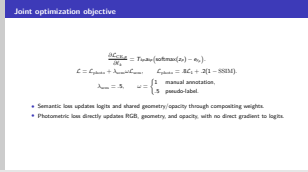
Joint optimization objective

$$\frac{\partial \mathcal{L}_{\mathrm{CE},p}}{\partial \ell_k} = T_{kp}\, a_{kp}\big(\mathrm{softmax}(z_p) - e_{y_p}\big).$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{photo}} + \lambda_{\mathrm{sem}}\omega\mathcal{L}_{\mathrm{sem}}, \qquad \mathcal{L}_{\mathrm{photo}} = .8\mathcal{L}_1 + .2(1 - \mathrm{SSIM}).$$

$$\lambda_{\mathrm{sem}} = .5, \qquad \omega = \begin{cases} 1 & \text{manual annotation,} \\ .5 & \text{pseudo-label.} \end{cases}$$

- Semantic loss updates logits and shared geometry/opacity through compositing weights.
- Photometric loss directly updates RGB, geometry, and opacity, with no direct gradient to logits.



Target: 0:25.

Training combines photometric and semantic objectives. The photometric term is 0.8 L1 plus 0.2 times one minus SSIM. Semantic cross-entropy has weight 0.5, and pseudo-label supervision receives an additional weight of 0.5. The semantic gradient updates logits directly and also affects shared geometry and opacity through the compositing weights.

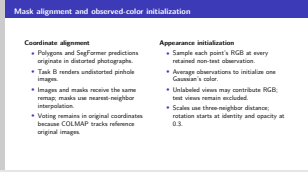
Mask alignment and observed-color initialization

Coordinate alignment

- Polygons and SegFormer predictions originate in distorted photographs.
- Task B renders undistorted pinhole images.
- Images and masks receive the same remap; masks use nearest-neighbor interpolation.
- Voting remains in original coordinates because COLMAP tracks reference original images.

Appearance initialization

- Sample each point's RGB at every retained non-test observation.
- Average observations to initialize one Gaussian's color.
- Unlabeled views may contribute RGB; test views remain excluded.
- Scales use three-neighbor distance; rotation starts at identity and opacity at 0.3.



Target: 0:20.

Two implementation details prevent coordinate and appearance errors. First, images and masks receive the same undistortion map, with nearest-neighbor interpolation for discrete labels. Second, each Gaussian color is initialized by averaging RGB samples from retained non-test observations. Voting remains in original coordinates because the supplied COLMAP tracks refer to the original photographs.

Task B experimental setup

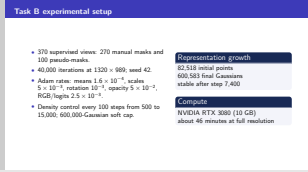
- 370 supervised views: 270 manual masks and 100 pseudo-masks.
- 40,000 iterations at 1320×989; seed 42.
- Adam rates: means 1.6×10^{-4}, scales 5×10^{-3}, rotation 10^{-3}, opacity 5×10^{-2}, RGB/logits 2.5×10^{-3}.
- Density control every 100 steps from 500 to 15,000; 600,000-Gaussian soft cap.

Representation growth

82,518 initial points
600,583 final Gaussians
stable after step 7,400

Compute

NVIDIA RTX 3080 (10 GB)
about 46 minutes at full resolution



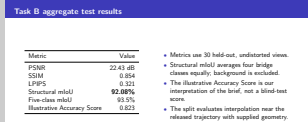
Target: 0:20.

Task B trains for 40,000 iterations at full image resolution with seed 42. Density control runs from step 500 to 15,000 with a soft cap near 600,000 Gaussians. The representation grows from 82,518 points to 600,583 Gaussians and stabilizes after step 7,400. Full-resolution training takes about 46 minutes on an RTX 3080.

Task B aggregate test results

Metric	Value
PSNR	22.43 dB
SSIM	0.854
LPIPS	0.321
Structural mIoU	**92.08%**
Five-class mIoU	93.5%
Illustrative Accuracy Score	0.823

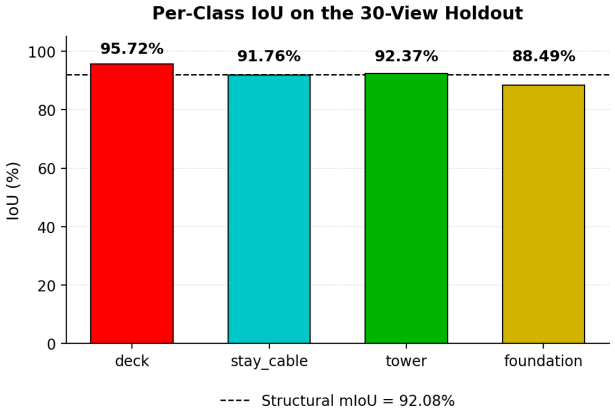
- Metrics use 30 held-out, undistorted views.
- Structural mIoU averages four bridge classes equally; background is excluded.
- The illustrative Accuracy Score is our interpretation of the brief, not a blind-test score.
- The split evaluates interpolation near the released trajectory with supplied geometry.



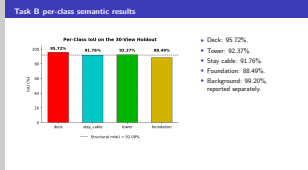
Target: 0:25.

On the 30 held-out views, the final model reaches 22.43 decibels PSNR, 0.854 SSIM, 0.321 LPIPS, and 92.08 percent structural mIoU. Background is excluded from structural mIoU, while the five-class average is 93.5 percent. The displayed Accuracy Score is an internal interpretation because the brief does not fully specify how to combine the visual metrics.

Task B per-class semantic results



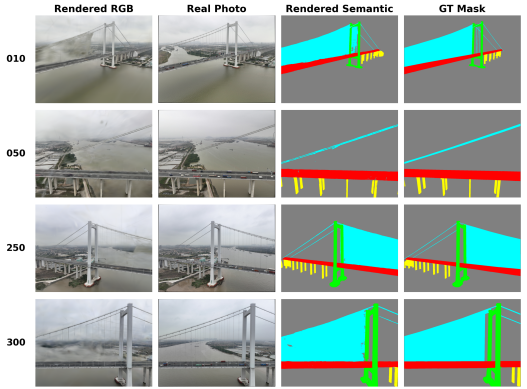
- Deck: 95.72%.
- Tower: 92.37%.
- Stay cable: 91.76%.
- Foundation: 88.49%.
- Background: 99.20%, reported separately.



Target: 0:20.

Per-class results remain strong across the four structural components. Deck is highest at 95.72 percent. Tower and stay cable both exceed 91 percent, and foundation reaches 88.49 percent. Background reaches 99.20 percent but is reported separately and does not dominate the structural mean.

Qualitative held-out views



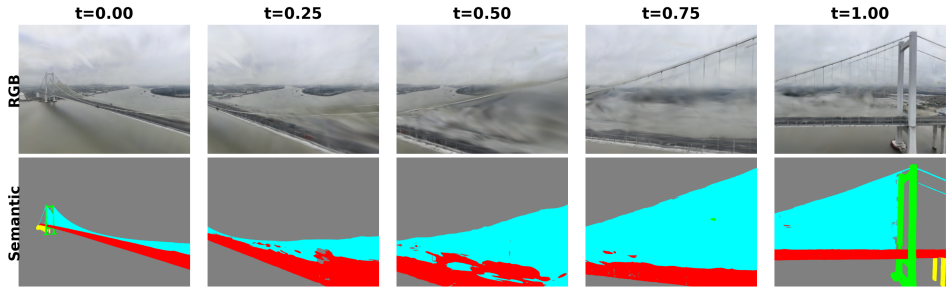
- View 250 is the cleanest RGB example; views 010 and 300 show visible blur and streaking.
- Semantic regions remain recognizable despite appearance artifacts, but four examples do not replace full-holdout statistics.



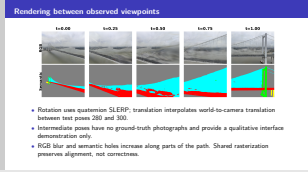
Target: 0:20.

These four held-out examples compare rendered RGB and semantics with ground truth. View 250 has the cleanest appearance. Views 010 and 300 show more blur and streaking, but their component regions remain recognizable. These examples illustrate behavior; the quantitative results still come from all 30 test views.

Rendering between observed viewpoints



- Rotation uses quaternion SLERP; translation interpolates world-to-camera translation between test poses 280 and 300.
- Intermediate poses have no ground-truth photographs and provide a qualitative interface demonstration only.
- RGB blur and semantic holes increase along parts of the path. Shared rasterization preserves alignment, not correctness.

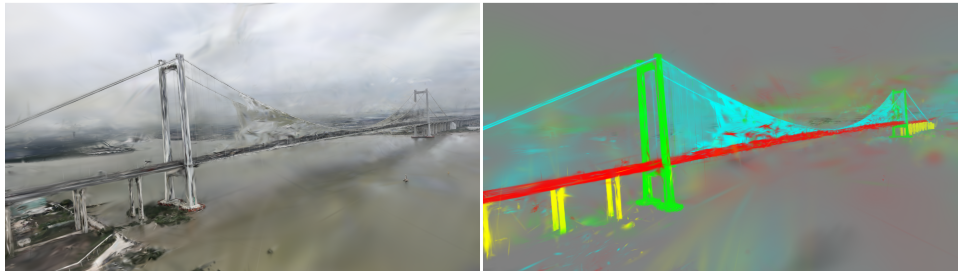


Target: 0:20.

We also render intermediate cameras between test poses 280 and 300. Rotation uses quaternion interpolation and translation is interpolated between the supplied poses. No ground-truth photographs exist for these intermediate views, so this is only a qualitative demonstration. Shared rendering maintains RGB-semantic alignment, although blur and semantic holes increase along the path.

Explicit RGB and semantic splats

(a) True-color splat render **(b) Colored by predicted semantic class**



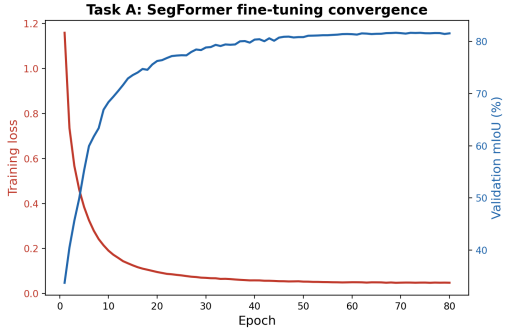
- One trained representation appears as learned RGB and per-Gaussian argmax class colors.
- Both screenshots use the same viewer camera pose, enabling direct spatial comparison of RGB appearance and per-Gaussian semantic colors.
- Per-Gaussian recoloring differs from evaluated semantic rendering, which composites logits before argmax.

Explicit RGB and semantic splats
(a) True-color splat render (b) Colored by predicted semantic class
• One trained representation appears as learned RGB and per-Gaussian argmax class colors.
• Both screenshots use the same viewer camera pose, enabling direct spatial comparison of RGB appearance and per-Gaussian semantic colors.
• Per-Gaussian recoloring differs from evaluated semantic rendering, which composites logits before argmax.

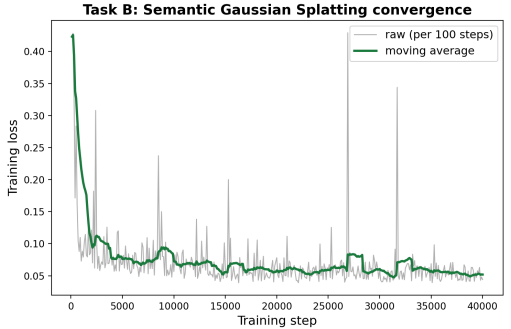
Target: 0:20.

These viewer screenshots show the same trained scene and exactly the same camera pose. The left panel uses learned RGB, while the right recolors each Gaussian by its predicted class. This helps inspect the 3D label distribution. It differs from the evaluated semantic image, where logits are composited first and argmax is applied afterward.

Training convergence

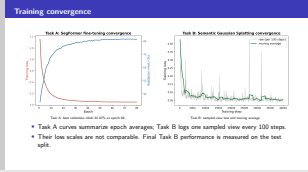


Task A: best validation mIoU 81.67% at epoch 69.



Task B: sampled-view loss and moving average.

- Task A curves summarize epoch averages; Task B logs one sampled view every 100 steps.
- Their loss scales are not comparable. Final Task B performance is measured on the test split.



Target: 0:15.

Task A converges smoothly and selects epoch 69. Task B shows noisier sampled-view loss because each logged point comes from one training view, but its moving average decreases and stabilizes. The two loss curves use different definitions and scales, so we do not compare their magnitudes.

Resolution ablation

Training	PSNR	SSIM	LPIPS	$\mathrm{mIoU}_{\mathrm{struct}}$
Half	22.03	.838	.330	89.71%
Full	**22.43**	**.854**	**.321**	**92.08%**

- Matched 40,000-step runs share the split, poses, initialization, losses, density schedule, and full-resolution evaluation.
- Full resolution improves PSNR by 0.40 dB, SSIM by 0.016, and structural mIoU by 2.37 points; LPIPS decreases by 0.009.
- Training time increases from about 18 to 46 minutes on the same RTX 3080.
- One run per resolution does not quantify random-seed uncertainty.

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Target: 0:20.

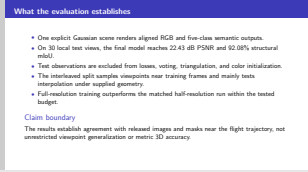
The resolution ablation uses matched 40,000-step runs and evaluates both checkpoints at full resolution. Full-resolution training improves PSNR by 0.40 decibels and structural mIoU by 2.37 points. SSIM also increases and LPIPS decreases. The cost is training time, which rises from about 18 to 46 minutes.

What the evaluation establishes

- One explicit Gaussian scene renders aligned RGB and five-class semantic outputs.
- On 30 local test views, the final model reaches 22.43 dB PSNR and 92.08% structural mIoU.
- Test observations are excluded from losses, voting, triangulation, and color initialization.
- The interleaved split samples viewpoints near training frames and mainly tests interpolation under supplied geometry.
- Full-resolution training outperforms the matched half-resolution run within the tested budget.

Claim boundary

The results establish agreement with released images and masks near the flight trajectory, not unrestricted viewpoint generalization or metric 3D accuracy.

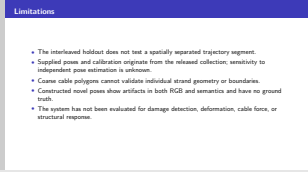


Target: 0:20.

The evaluation supports three conclusions. One explicit scene produces aligned RGB and component labels. The local test result reaches 22.43 decibels PSNR and 92.08 percent structural mIoU. Full resolution performs better than the matched half-resolution run. Because the split interleaves nearby viewpoints, these results mainly establish interpolation near the released flight trajectory.

Limitations

- The interleaved holdout does not test a spatially separated trajectory segment.
- Supplied poses and calibration originate from the released collection; sensitivity to independent pose estimation is unknown.
- Coarse cable polygons cannot validate individual strand geometry or boundaries.
- Constructed novel poses show artifacts in both RGB and semantics and have no ground truth.
- The system has not been evaluated for damage detection, deformation, cable force, or structural response.



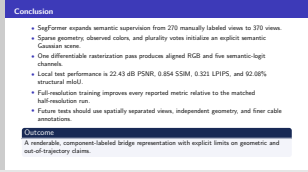
Target: 0:15.
The main limitations are clear. The holdout is not a spatially separate flight segment, and all evaluation uses supplied camera geometry. Coarse cable polygons cannot validate individual strands. Constructed views have no ground truth, and the system has not been tested for damage detection, deformation, cable force, or structural response.

Conclusion

- SegFormer expands semantic supervision from 270 manually labeled views to 370 views.
- Sparse geometry, observed colors, and plurality votes initialize an explicit semantic Gaussian scene.
- One differentiable rasterization pass produces aligned RGB and five semantic-logit channels.
- Local test performance is 22.43 dB PSNR, 0.854 SSIM, 0.321 LPIPS, and 92.08% structural mIoU.
- Full-resolution training improves every reported metric relative to the matched half-resolution run.
- Future tests should use spatially separated views, independent geometry, and finer cable annotations.

Outcome

A renderable, component-labeled bridge representation with explicit limits on geometric and out-of-trajectory claims.

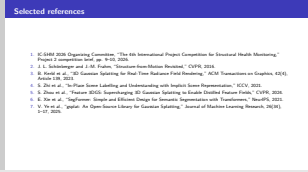


Target: 0:20.

In conclusion, SegFormer expands supervision from 270 manual masks to 370 views. Sparse geometry, observed colors, and semantic voting initialize one explicit Gaussian scene. A single differentiable rendering pass then produces aligned RGB and five-class semantics. The method performs strongly on the local holdout, while future work should test separated viewpoints, independent geometry, and finer cable labels.

Selected references

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3. B. Kerbl et al., "3D Gaussian Splatting for Real-Time Radiance Field Rendering," ACM Transactions on Graphics, 42(4), Article 139, 2023.
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Target: 0:10.

These are the main references behind the reconstruction, segmentation, semantic scene representation, and rendering components. Thank you for your attention.